

# CUSTOMER CHURN PREDICTION AND EXPLAINABILITY DASHBOARD

Predictive Analytics and SHAP Feature Attribution Analysis

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Domain: Predictive Machine Learning

Stack: XGBoost, SHAP, Streamlit

## PERFORMANCE METRICS

|                                          |                                                |                                               |                                               |
|------------------------------------------|------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| <div>0.94</div> <div>AUC ROC Score</div> | <div>88.0%</div> <div>Minority Precision</div> | <div>70,000+</div> <div>Telecom Records</div> | <div>25+</div> <div>Engineered Features</div> |
|------------------------------------------|------------------------------------------------|-----------------------------------------------|-----------------------------------------------|

## FEATURE IMPORTANCE RANKING

| Feature Name    | Importance Weight | Business Impact                                      |
|-----------------|-------------------|------------------------------------------------------|
| tenure_months   | 38.65%            | Longer tenure reduces churn risk significantly       |
| support_tickets | 32.74%            | High customer service tickets signal escalation risk |
| payment_delays  | 24.42%            | Frequent billing delays correlate with cancellation  |
| monthly_charges | 4.19%             | Higher pricing tiers impact price sensitive segments |